Formalization of the Meta-Audit Protocol: Who Audits the Auditor

SCX Equality Framework — Reflexivity Verification Chapter

SCX

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Abstract

The SCX Equality Framework's core operational principle is the Audit Sword, but can the audit itself be trusted? This paper rigorously formalizes the Meta-Audit protocol, addressing the reflexive question "who audits the auditor." Core contributions include: (1) formal definition of the maintainer bias parameter g, modeling systematic auditor deviation as a detectable statistic; (2) a bias detection theorem based on Hoeffding's inequality, giving a lower bound on the number of independent replications required to detect $g \neq 0$ at a given confidence level; (3) formalization of multi-maintainer rotation and consensus mechanisms, including k-of-n consensus protocols, rotation scheduling optimization, and exponential decay bounds on consensus failure probability. All theorems are rigorously proved and checked for consistency with the core SCX axiom (Theorem 3: noise-difficulty indistinguishability).

Keywords: Meta-audit, Hoeffding bound, maintainer rotation, consensus protocol, SCX framework, reflexivity, statistical detection, auditor bias.

1 Introduction

1.1 Motivation: The Auditor's Reflexivity Dilemma

The SCX Equality Framework ensures transparency and verifiability of all agent evaluation and evolution processes through the Audit Sword mechanism. Each auditor (Maintainer) is responsible for computing and verifying the Cercis Score $S(x) = Q(x) + \eta \cdot N(x)$, adjudicating the quality–novelty decomposition. However, a fundamental reflexive question arises: **if the auditor itself harbors systematic bias, who audits the auditor?**

This dilemma is not unique to SCX. In law, scientific peer review, financial auditing—all human institutions—the auditor-bias problem exists in some form: juries have prejudice, reviewers favor certain methodologies, auditors are captured by interests. SCX's uniqueness lies in the algorithmic nature of its audit process: because the audit basis (Cercis Score, Situs encoding, gradient-field diagnostics) is mathematically defined, bias detection against the auditor can be rigorously formalized rather than relying solely on institutional trust.

1.2 Core Questions and Contributions

This paper focuses on three core questions:

Q1 Bias Detectability. How is the systematic bias parameter g_i of maintainer M_i defined? How can $g_i \neq 0$ be detected using public audit logs and third-party random replications via statistical tests?

Q2 Detection Sample Complexity. Given target confidence level $1 - \delta$ and minimum detectable bias ε, how many independent replications are needed? What guarantees does the Hoeffding bound provide?

Q3 Structural Defenses. How are multi-maintainer rotation and consensus mechanisms formalized? Under partially malicious (Byzantine) maintainers, how is consensus reliability guaranteed?

1.3 Consistency with SCX Core Axioms

The meta-audit protocol must remain consistent with Theorem 3 (noise-difficulty indistinguishability). Theorem 3 states that label noise and intrinsic difficulty cannot be distinguished from observational data. Meta-audit introduces a new layer—the systematic bias g of the auditor. The critical question: is g subject to Theorem 3's indistinguishability constraint? This paper proves: **through cross-sectional replication rather than longitudinal inference, the detection of g bypasses Theorem 3's barrier**, because detection relies on agreement/disagreement patterns across different auditors on the same input, rather than intrinsic analysis of a single auditor's output.

2 Formal Framework

2.1 Audit Model

Definition 2.1 (Audit Instance)**.** *An audit instance is a triple* $(x, S, \mathcal{A})$*, where:*

- $x \in \mathcal{X}$*: the input being audited (e.g., a paper, a model output, a decision record);*
- $S : \mathcal{X} \to \mathbb{R}$*: the true Cercis Score function (defined by SCX axioms,* $S(x) = Q(x)+\eta N(x)$*);*
- $\mathcal{A} = \{A_1, \ldots, A_m\}$*: the set of auditors (maintainers).*

Auditor A_i *outputs an audit verdict* $\hat{S}_i(x)$ *as an estimate of* $S(x)$*.*

Definition 2.2 (Maintainer Bias Model)**.** *Maintainer* M_i *(viewed as auditor* A_i*) produces output* $\hat{S}_i(x)$ *following:*

$$\hat{S}_i(x) = S(x) + g_i + \epsilon_i(x), \tag{1}$$

where:

- $g_i \in \mathbb{R}$*: maintainer* i*'s* ***systematic bias*** *(meta-bias).* $g_i = 0$ *indicates an unbiased maintainer,* $g_i > 0$ *indicates systematic overestimation,* $g_i < 0$ *indicates systematic underestimation;*
- $\epsilon_i(x) \sim \mathcal{D}_i(0, \sigma_i^2)$*: zero-mean random error term,* $\mathbb{E}[\epsilon_i(x)] = 0$*,* $\mathrm{Var}[\epsilon_i(x)] = \sigma_i^2$*;*
- $\epsilon_i(x)$ *is independent across different* x *and different* i*.*

Remark 2.3. *Model* (**??**) *distinguishes two types of audit error: (1)* g_i*—systematic bias attributable to maintainer identity (ideological capture, incentive misalignment); (2)* $\epsilon_i(x)$*—random noise (bounded rationality, occasional errors, input ambiguity). The core objective of meta-audit is* ***detecting*** $g_i \neq 0$*, not estimating* $\epsilon_i(x)$*.*

Definition 2.4 (Audit Log)**.** *Maintainer* M_i*'s public audit log* $\mathcal{L}_i$ *is a publicly verifiable sequence of records:*

$$\mathcal{L}_i = \{(x_j, \hat{S}_i(x_j), t_j, \mathsf{hash}_j)\}_{j=1}^{n_i}, \tag{2}$$

where t_j *is a timestamp, and* $\mathsf{hash}_j = H(x_j \| \hat{S}_i(x_j) \| t_j \| \mathsf{hash}_{j-1})$ *guarantees tamper-evident (append-only Merkle chain) logging.*

2.2 Statistical Framework for Bias Detection

Definition 2.5 (Pairwise Discrepancy). *For the same input x, the output difference between two independent maintainers M_i and M_j is:*

$$\Delta_{ij}(x) = \hat{S}_i(x) - \hat{S}_j(x) = (g_i - g_j) + (\epsilon_i(x) - \epsilon_j(x)). \tag{3}$$

Its expectation is:

$$\mathbb{E}[\Delta_{ij}(x)] = g_i - g_j. \tag{4}$$

Definition 2.6 (Third-Party Replication). *A Third-Party Replicator (TPR) is an independent entity that randomly draws n samples $\{x_1, \ldots, x_n\} \sim \mathcal{X}$ from the audited input space, independently computes $\hat{S}_{TPR}(x_k)$, and compares it against the corresponding records in maintainer M_i's public log $\mathcal{L}_i$.*

Definition 2.7 (Replication Discrepancy Statistic). *Given n independent replication samples, define the replication discrepancy statistic:*

$$D_n^{(i)} = \frac{1}{n} \sum_{k=1}^{n} \left(\hat{S}_i(x_k) - \hat{S}_{TPR}(x_k) \right). \tag{5}$$

*Under Model (**??**), assuming the TPR is unbiased ($g_{TPR} = 0$ or known and calibrated):*

$$\mathbb{E}[D_n^{(i)}] = g_i, \quad \mathrm{Var}[D_n^{(i)}] = \frac{\sigma_i^2 + \sigma_{TPR}^2}{n}. \tag{6}$$

3 Bias Detection Theorems via Hoeffding Bound

3.1 Hoeffding's Inequality Review

Lemma 3.1 (Hoeffding's Inequality). *Let $X_1, \ldots, X_n$ be independent random variables with $X_k \in [a_k, b_k]$ almost surely. Let $\bar{X} = \frac{1}{n}\sum_{k=1}^{n} X_k$. Then for any $t > 0$:*

$$\mathbb{P}\left(|\bar{X} - \mathbb{E}[\bar{X}]| \geq t\right) \leq 2\exp\left(-\frac{2n^2t^2}{\sum_{k=1}^{n}(b_k - a_k)^2}\right). \tag{7}$$

In particular, if all $X_k \in [a, b]$, then:

$$\mathbb{P}\left(|\bar{X} - \mathbb{E}[\bar{X}]| \geq t\right) \leq 2\exp\left(-\frac{2nt^2}{(b-a)^2}\right). \tag{8}$$

3.2 Bounded Audit Assumption

Definition 3.2 (Bounded Audit Output). *The maintainer's audit output $\hat{S}_i(x)$ and TPR's audit output $\hat{S}_{TPR}(x)$ satisfy:*

$$\hat{S}_i(x), \hat{S}_{TPR}(x) \in [S_{\min}, S_{\max}], \tag{9}$$

where $S_{\min}, S_{\max}$ are the global bounds of the Cercis Score (determined by the domains of Q and N). Define the audit range width $R = S_{\max} - S_{\min}$.

Remark 3.3. *In the SCX framework, $Q(x)$ is typically normalized to $[0,1]$, $N(x)$ to $[0,1]$, hence $S(x) \in [0, 1+\eta]$. For a given η, $R = 1 + \eta$ is a known constant.*

3.3 Main Theorem: Hoeffding Bias Detection

Theorem 3.4 (Hoeffding Bias Detection Theorem)**.** *Let maintainer M_i have systematic bias g_i. Assume:*

(A1) Audit output is bounded: $\hat{S}_i(x), \hat{S}_{TPR}(x) \in [S_{\min}, S_{\max}]$, $R = S_{\max} - S_{\min}$;

(A2) TPR is unbiased: $g_{TPR} = 0$ (or known bias g_{TPR} is calibrated);

(A3) TPR independently draws n samples $\{x_k\}_{k=1}^n \overset{i.i.d.}{\sim} \mathcal{X}$ and independently computes audits.

Define detection threshold $\tau > 0$. Then:

(i) ***False Positive Control.*** *If $g_i = 0$ (M_i is unbiased), then:*

$$\mathbb{P}\left(|D_n^{(i)}| \geq \tau \;\middle|\; g_i = 0\right) \leq 2\exp\left(-\frac{2n\tau^2}{R^2}\right). \tag{10}$$

(ii) ***Detection Power.*** *If $|g_i| \geq \varepsilon > 0$, then:*

$$\mathbb{P}\left(|D_n^{(i)}| \geq \tau \;\middle|\; |g_i| \geq \varepsilon\right) \geq 1 - 2\exp\left(-\frac{2n(\varepsilon-\tau)^2}{R^2}\right). \tag{11}$$

(iii) ***Sample Complexity.*** *To detect $|g_i| \geq \varepsilon$ at significance level α (i.e., $\mathbb{P}$(False Positive) $\leq \alpha$) and power $1-\beta$ (i.e., $\mathbb{P}$(detection) $\geq 1-\beta$), the required sample size satisfies:*

$$n \geq \frac{R^2}{2} \cdot \frac{\left(\sqrt{\log(2/\alpha)} + \sqrt{\log(2/\beta)}\right)^2}{\varepsilon^2}. \tag{12}$$

Proof. **(i) False Positive Control.** Under $g_i = 0$, $\mathbb{E}[D_n^{(i)}] = 0$. Define $Y_k = \hat{S}_i(x_k) - \hat{S}_{\text{TPR}}(x_k)$. Since both terms lie in $[S_{\min}, S_{\max}]$, $Y_k \in [-R, R]$. By Hoeffding's inequality (**??**), for n independent Y_k:

$$\mathbb{P}\left(|D_n^{(i)} - 0| \geq \tau\right) \leq 2\exp\left(-\frac{2n\tau^2}{(2R)^2}\right) = 2\exp\left(-\frac{n\tau^2}{2R^2}\right).$$

Note: the denominator in the original Hoeffding bound is $\sum(b_k - a_k)^2 = n \cdot (2R)^2 = 4nR^2$, yielding $\exp(-2n^2\tau^2/4nR^2) = \exp(-n\tau^2/2R^2)$. A tighter form considers the bound of Y_k as $[-R, R]$ with range $2R$, substituting into (**??**):

$$\mathbb{P}\left(|\bar{Y} - \mathbb{E}[\bar{Y}]| \geq \tau\right) \leq 2\exp\left(-\frac{2n\tau^2}{(2R)^2}\right) = 2\exp\left(-\frac{n\tau^2}{2R^2}\right).$$

This bound is correct. More rigorously: since $Y_k = (\hat{S}_i(x_k) - S_{\min}) - (\hat{S}_{\text{TPR}}(x_k) - S_{\min})$ and each term is in $[0, R]$, $Y_k \in [-R, R]$. Let $Z_k = (Y_k + R)/(2R) \in [0, 1]$, then $\bar{Z} = (\bar{Y} + R)/(2R)$, $\mathbb{E}[\bar{Z}] = R/(2R) = 1/2$ (when $g_i = 0$). Hence:

$$\mathbb{P}(|\bar{Y}| \geq \tau) = \mathbb{P}(|\bar{Z} - 1/2| \geq \tau/(2R)) \leq 2\exp(-2n(\tau/(2R))^2) = 2\exp(-n\tau^2/(2R^2)).$$

This gives the precise derivation of (**??**). $\square$

(ii) Detection Power. When $|g_i| \geq \varepsilon$, $\mathbb{E}[D_n^{(i)}] = g_i$. Without loss of generality, assume $g_i \geq \varepsilon > 0$ (the negative case is symmetric). Then:

$$\begin{aligned}
\mathbb{P}\left(|D_n^{(i)}| \geq \tau\right) &\geq \mathbb{P}\left(D_n^{(i)} \geq \tau\right) \\
&= \mathbb{P}\left(D_n^{(i)} - g_i \geq \tau - g_i\right) \\
&\geq \mathbb{P}\left(D_n^{(i)} - g_i \geq \tau - \varepsilon\right) \quad \text{(since } g_i \geq \varepsilon\text{)} \\
&= 1 - \mathbb{P}\left(D_n^{(i)} - g_i < \tau - \varepsilon\right) \\
&\geq 1 - \mathbb{P}\left(|D_n^{(i)} - g_i| \geq \varepsilon - \tau\right) \quad \text{(when } \varepsilon > \tau\text{)} \\
&\geq 1 - 2\exp\left(-\frac{2n(\varepsilon - \tau)^2}{R^2}\right).
\end{aligned}$$

We used Hoeffding: $\mathbb{P}(|D_n^{(i)} - \mathbb{E}[D_n^{(i)}]| \geq \varepsilon - \tau) \leq 2\exp(-2n(\varepsilon - \tau)^2/R^2)$. □

(iii) Sample Complexity. From (i), controlling false positive rate $\leq \alpha$ requires:

$$2\exp\left(-\frac{2n\tau^2}{R^2}\right) \leq \alpha \implies n \geq \frac{R^2}{2\tau^2}\log\frac{2}{\alpha}.$$

From (ii), controlling type II error rate $\leq \beta$ (power $\geq 1 - \beta$) requires:

$$2\exp\left(-\frac{2n(\varepsilon - \tau)^2}{R^2}\right) \leq \beta \implies n \geq \frac{R^2}{2(\varepsilon - \tau)^2}\log\frac{2}{\beta}.$$

The optimal τ simultaneously satisfies both; choosing $\tau = \varepsilon/2$ (symmetric allocation):

$$n \geq \max\left\{\frac{R^2}{2(\varepsilon/2)^2}\log\frac{2}{\alpha}, \frac{R^2}{2(\varepsilon/2)^2}\log\frac{2}{\beta}\right\} = \frac{2R^2}{\varepsilon^2}\max\left\{\log\frac{2}{\alpha}, \log\frac{2}{\beta}\right\}.$$

A tighter analysis (jointly minimizing n) yields (**??**). □ □

Remark 3.5 (Improved Constants)**.** *Using the Bernstein inequality (which exploits variance information), the sample complexity can be further improved to:*

$$n \geq \frac{2(\sigma_i^2 + \sigma_{TPR}^2)\log(2/\alpha) + \frac{2R}{3}\varepsilon\log(2/\alpha)}{\varepsilon^2},$$

*which is significantly better than the Hoeffding bound when $\sigma_i^2 + \sigma_{TPR}^2 \ll R^2$. However, the advantage of the Hoeffding bound is that it **does not depend on variance estimates** and is valid for any bounded distribution—in the meta-audit setting, σ_i^2 itself may be manipulated by a malicious maintainer, so distribution-free bounds are more robust.*

3.4 Multi-Replicator Joint Detection

Theorem 3.6 (Multi-TPR Joint Detection Theorem)**.** *Let there be K independent third-party replicators $TPR_1, \ldots, TPR_K$, each independently computing n audit samples with maintainer M_i. Define the aggregate statistic:*

$$\bar{D}_{n,K}^{(i)} = \frac{1}{K}\sum_{r=1}^{K} D_n^{(i,r)}, \tag{13}$$

where $D_n^{(i,r)}$ is the replication discrepancy between M_i and TPR_r.

If $\mathbb{E}[D_n^{(i,r)}] = g_i$ for all r (all TPRs are unbiased and consistently calibrated), then:

$$\mathbb{P}\left(|\bar{D}_{n,K}^{(i)} - g_i| \geq t\right) \leq 2\exp\left(-\frac{2nKt^2}{R^2}\right). \tag{14}$$

Proof. $\bar{D}_{n,K}^{(i)}$ is the mean of nK independent observations (the n samples per TPR are independent, and the K TPRs are independent, giving nK independent observations total), each in $[-R, R]$. Directly apply Hoeffding's inequality. □ □

Corollary 3.7 (Replicator Crowdsourcing)**.** *By increasing the number of replicators K, detection precision can be improved without increasing the burden on any individual replicator. The effective sample size is $n_{\mathit{eff}} = n \cdot K$. In particular, if each TPR audits only $n = 1$ sample, the joint detection power of K TPRs is equivalent to a single TPR auditing K samples (under the independence and unbiasedness assumptions).*

4 Maintainer Rotation Mechanism

4.1 Necessity of Rotation

Even when $g_i \neq 0$ is detected, ex-post punishment (e.g., revocation of maintainer status) has a time lag. More importantly, certain bias patterns may require long periods to accumulate to statistical significance—this is called a **slow poisoning attack**: the maintainer gradually introduces bias at an extremely low rate $g_i^{(t)} = \delta \cdot t/T$, such that no fixed-window detection statistic is significant, but the long-term cumulative effect is substantial.

Rotation provides an ex-ante defense: by limiting the continuous tenure of any single maintainer, the maximum cumulative bias is bounded.

4.2 Rotation Scheduling Model

Definition 4.1 (Maintainer Rotation Schedule)**.** *Let the maintainer pool be $\mathcal{M} = \{M_1, \ldots, M_N\}$, $N \geq 2$. Time is discretized into epochs $t = 1, 2, \ldots, T$. In each epoch t, the set of active maintainers is $\mathcal{A}_t \subseteq \mathcal{M}$ of size $m_t = |\mathcal{A}_t|$ (typically $m_t = 1$ or $m_t = 3$).*

A rotation schedule $\mathcal{R}$ specifies the sequence of $\mathcal{A}_t$, satisfying:

(C1) ***Coverage:*** $\bigcup_{t=1}^{T} \mathcal{A}_t = \mathcal{M}$*, every maintainer has an opportunity to participate;*

(C2) ***Term Limit:*** *$\forall i$, the number of consecutive active epochs $\leq L_{\max}$ (maximum consecutive tenure);*

(C3) ***Cool-down:*** *$\forall i$, after leaving office, at least $C_{\min}$ epochs must pass before reappointment (minimum cool-down period).*

Theorem 4.2 (Rotation Bias Upper Bound)**.** *Under rotation schedule $\mathcal{R}$, if each maintainer M_i's bias g_i satisfies $|g_i| \leq G_{\max}$, then over T epochs, the cumulative bias along any audit path satisfies:*

$$\left| \sum_{t=1}^{T} g_{a(t)} \right| \leq G_{\max} \cdot \min\left(T, N \cdot L_{\max} + (N-1) \cdot \left\lceil \frac{T}{L_{\max} + C_{\min}} \right\rceil \cdot L_{\max} \right), \tag{15}$$

where $a(t)$ is the index of the active maintainer in epoch t. When T is sufficiently large, the cumulative bias upper bound is $O(G_{\max} \cdot T \cdot \frac{L_{\max}}{L_{\max}+C_{\min}})$, linear in T but with slope $G_{\max} \cdot \frac{L_{\max}}{L_{\max}+C_{\min}} < G_{\max}$.

Proof. Worst case: the most biased maintainer ($|g_i| = G_{\max}$) occupies as many epochs as possible. Under the rotation constraint, this maintainer can be active for at most $L_{\max}$ epochs out of every $L_{\max} + C_{\min}$ epochs. Hence the maximum number of active epochs out of T is $\lceil T/(L_{\max} + C_{\min}) \rceil \cdot L_{\max}$. Also accounting for the pool size constraint N (must rotate to others), take the smaller upper bound. □ □

Corollary 4.3 (Optimal Rotation Parameters). *To minimize cumulative bias, set $L_{\max}$ as small as possible and $C_{\min}$ as large as possible. But practical constraints (continuity of maintenance, retention of expertise) require $L_{\max} \geq 1$ and $L_{\max} + C_{\min}$ not exceeding the rotation cycle of the maintainer pool. The optimal practical choice: $L_{\max} = 1$ (rotate every epoch), $C_{\min} = N - 1$ (all maintainers complete one full cycle).*

5 Multi-Maintainer Consensus

5.1 Necessity of Consensus

While individual third-party replication detection is effective, it relies on the TPR's unbiasedness assumption—this is the recursive nature of the meta-audit reflexivity problem: what if the TPR itself is biased? Multi-maintainer consensus mechanisms solve this through **cross-sectional agreement tests**: even if individual maintainers may be biased, majority consensus can recover the true audit result under appropriate conditions.

5.2 k-of-n Consensus Protocol

Definition 5.1 (k-of-n Consensus). *Given n maintainers producing audit verdicts $\hat{S}_1(x), \ldots, \hat{S}_n(x)$ for the same input x, the k-of-n consensus rule is:*

$$\hat{S}_{cons}(x) = median_k(\hat{S}_1(x), \ldots, \hat{S}_n(x)), \tag{16}$$

where $median_k$ denotes selecting the k-th order statistic from the sorted values ($1 \leq k \leq n$). Standard majority consensus corresponds to $k = \lceil n/2 \rceil$ (the median).

Definition 5.2 (Byzantine Maintainer Model). *Among n maintainers, b are **Byzantine** (can deviate arbitrarily, including strategic output to disrupt consensus), and $n - b$ are **honest and unbiased** ($g_i = 0$, $\epsilon_i(x)$ i.i.d.). The output distribution of honest maintainers is $\hat{S}_i(x) \sim \mathcal{F}$, where $\mathcal{F}$ is symmetric about $S(x)$ with bounded support $[S_{\min}, S_{\max}]$.*

Theorem 5.3 (Byzantine Consensus Reliability). *Under k-of-n consensus, with b Byzantine maintainers and $n - b$ honest maintainers, if:*

$$b < \min(k, n - k + 1), \tag{17}$$

then the consensus output $\hat{S}_{cons}(x)$ lies within the range of honest maintainer outputs:

$$\min_{i \in \mathcal{H}} \hat{S}_i(x) \leq \hat{S}_{cons}(x) \leq \max_{i \in \mathcal{H}} \hat{S}_i(x), \tag{18}$$

where $\mathcal{H}$ is the set of honest maintainers ($|\mathcal{H}| = n - b$).

In particular, for median consensus ($k = \lceil n/2 \rceil$), the condition reduces to $b < n/2$ (the classic Byzantine fault tolerance upper bound), and:

$$\mathbb{P}\left(|\hat{S}_{cons}(x) - S(x)| \geq t\right) \leq 2 \exp\left(-\frac{2(n-b)t^2}{R^2}\right). \tag{19}$$

Proof. The condition $b < \min(k, n - k + 1)$ ensures that among the n sorted outputs, regardless of where Byzantine maintainers place their outputs, the k-th order statistic necessarily falls within the interval of honest maintainer outputs (since $b < k$ means at least one honest output is among the first k, and $b < n - k + 1$ means at least one honest output is among the last $n - k + 1$, combined implying the k-th lies in the honest interval).

For the median case: $k = \lceil n/2 \rceil$, $\min(k, n - k + 1) = \lceil n/2 \rceil$ (when n is odd) or $n/2$ (when n is even). Condition $b < n/2$ means honest maintainers form a strict majority. The median then falls within the range of honest outputs. Furthermore, since the median of honest outputs is a sample median of honest outputs, we can apply the Hoeffding bound to obtain (**??**). Note: the median does not use all information from the n honest maintainers; convergence rate is $O(1/\sqrt{n-b})$. Tighter bounds can use large deviation theory for order statistics. □ □

Remark 5.4 (Relationship to Classical Distributed Systems Literature)**.** *Classical Byzantine fault tolerance (e.g., PBFT) requires $n > 3b$ to guarantee safety + liveness. The consensus in this paper only requires safety (audit results are trustworthy), not liveness (the audit process can wait until sufficient consistent opinions are collected). Hence the condition is looser: $n > 2b$. This corresponds to* ***static*** *consensus in asynchronous networks, consistent with Nakamoto consensus in blockchains (probabilistic, honest majority assumption $n > 2b$).*

5.3 Consensus Failure Probability

Theorem 5.5 (Hoeffding Weighted Consensus Bound)**.** *Consider weighted consensus:*

$$\hat{S}_{\textit{w-cons}}(x) = \sum_{i=1}^{n} w_i \hat{S}_i(x), \quad \sum_{i=1}^{n} w_i = 1, \quad w_i \geq 0. \tag{20}$$

The weights w_i are based on maintainers' historical audit accuracy (updated by meta-audit detection results). Assume the subset of honest maintainers $\mathcal{H}$ satisfies $\sum_{i\in\mathcal{H}} w_i = W_{\mathcal{H}} \in (0,1]$. Then:

$$\mathbb{P}\left(|\hat{S}_{\textit{w-cons}}(x) - S(x)| \geq t\right) \leq 2\exp\left(-\frac{2t^2}{R^2\sum_{i=1}^{n} w_i^2}\right) + \mathbb{P}(\textit{Byzantine deviation} > t\cdot(1 - W_{\mathcal{H}})). \tag{21}$$

Proof. Decompose the bias of weighted consensus:

$$\begin{aligned} |\hat{S}_{\text{w-cons}}(x) - S(x)| &= \left|\sum_{i\in\mathcal{H}} w_i(\hat{S}_i(x) - S(x)) + \sum_{i\notin\mathcal{H}} w_i(\hat{S}_i(x) - S(x))\right| \\ &\leq \underbrace{\left|\sum_{i\in\mathcal{H}} w_i(\hat{S}_i(x) - S(x))\right|}_{\text{honest maintainer contribution}} + \underbrace{\sum_{i\notin\mathcal{H}} w_i\cdot R}_{\text{Byzantine contribution}}. \end{aligned}$$

Honest maintainer contribution: $\sum_{i\in\mathcal{H}} w_i(\hat{S}_i(x) - S(x))$ is a weighted sum of zero-mean independent terms (note: $\hat{S}_i(x) - S(x)$ are independent across different i). View each $\hat{S}_i(x) - S(x)$ as a random variable bounded in $[-R, R]$, with weight w_i. By the weighted version of Hoeffding's inequality:

$$\mathbb{P}\left(\left|\sum_{i\in\mathcal{H}} w_i(\hat{S}_i(x) - S(x))\right| \geq t\right) \leq 2\exp\left(-\frac{2t^2}{\sum_{i\in\mathcal{H}} w_i^2\cdot(2R)^2}\right) = 2\exp\left(-\frac{t^2}{2R^2\sum_{i\in\mathcal{H}} w_i^2}\right).$$

A simpler bound: use $R^2\sum_i w_i^2$ as the denominator (including weights of all n maintainers, $\sum_{i\in\mathcal{H}} w_i^2 \leq \sum_{i=1}^{n} w_i^2$). □ □

6 The Meta-Audit Protocol

6.1 Protocol Description

Protocol 6.1 (Meta-Audit Protocol MetaAudit)**. Initialization.**

I1 Determine the maintainer pool $\mathcal{M} = \{M_1, \ldots, M_N\}$, $N \geq 3$.

I2 Set parameters: significance level α, minimum detectable bias ε, target power $1-\beta$, audit range $R = S_{\max} - S_{\min}$.

I3 Compute the required replication sample size n^* according to (**??**) from **??**.

I4 Initialize maintainer trust weights $\mathbf{w}^{(0)} = (1/N, \ldots, 1/N)$.

Per-Epoch Operation. For each epoch $t = 1, 2, \ldots$:

E1 Rotation Selection. Select the active maintainer set $\mathcal{A}_t$ according to rotation schedule $\mathcal{R}$.

E2 Audit Execution. Maintainers in $\mathcal{A}_t$ produce audit verdicts for the current audit batch $\mathcal{X}_t$ and append them to their respective logs $\mathcal{L}_i$.

E3 Consensus Aggregation. If $|\mathcal{A}_t| \geq 3$, execute k-of-n consensus ($k = \lceil |\mathcal{A}_t|/2 \rceil$) to obtain $\hat{S}_{\text{cons}}$. If $|\mathcal{A}_t| = 1$, use that maintainer's output directly.

E4 TPR Replication. Randomly select n^* previously audited $x \in \bigcup_{\tau \leq t} \mathcal{X}_\tau$, and have $K \geq 1$ third-party replicators independently re-audit them. Compute the replication discrepancy statistic $D_{n^*}^{(i,t)}$ for each $M_i \in \mathcal{A}_t$.

E5 Bias Test. For each M_i, perform the hypothesis test:

$$H_0 : g_i = 0 \quad \text{vs} \quad H_1 : |g_i| \geq \varepsilon.$$

If $|D_{n^*}^{(i,t)}| \geq \tau$ ($\tau = \varepsilon/2$), reject H_0—flag M_i as having significant bias.

E6 Weight Update. Update trust weights based on detection results:

$$w_i^{(t)} \propto w_i^{(t-1)} \cdot \exp\left(-\gamma \cdot \mathbf{1}_{|D_{n^*}^{(i,t)}| \geq \tau}\right), \tag{22}$$

then normalize so that $\sum_i w_i^{(t)} = 1$. The parameter $\gamma > 0$ controls penalty strength.

E7 Consensus Weight Adjustment. Use the updated $\mathbf{w}^{(t)}$ as the weighted consensus weights for the next epoch (if using weighted consensus).

E8 Maintainer Replacement. If M_i is flagged for bias in C_{strike} consecutive epochs, remove M_i from $\mathcal{M}$ and replace with a new maintainer.

6.2 Protocol Correctness

Theorem 6.2 (Meta-Audit Protocol Reliability)**.** *Under the* MetaAudit *protocol, assume:*

(H1) Audit outputs are bounded in $[S_{\min}, S_{\max}]$*;*

(H2) TPR replicators are mutually independent and independent of maintainers (can come from different institutional ecosystems);

(H3) At least $n - b > n/2$ *maintainers are honest and unbiased (Byzantine condition);*

(H4) The rotation schedule ensures each maintainer is tested by replication at least once every T_{cycle} *epochs.*

Then:

(a) ***Bias Detection Delay.*** *A malicious maintainer with bias* $|g_i| \geq \varepsilon$ *will be detected within* $O(T_{cycle} \cdot \log(2/\beta))$ *epochs with probability* $\geq 1 - \beta$*;*

(b) ***Consensus Error.*** *The weighted consensus output satisfies: for any $t > 0$,*

$$\mathbb{P}(|\hat{S}_{w\text{-}cons}(x) - S(x)| \geq t) \leq 2\exp\left(-\frac{2t^2}{R^2 \cdot \sum_{i\in\mathcal{H}} w_i^2}\right) + O(e^{-\gamma C_{strike}});$$

(c) ***Reflexive Consistency.*** *The protocol is consistent with SCX Theorem 3:* MetaAudit*'s bias detection does not rely on breaking noise/difficulty indistinguishability, because detection comes from cross-sectional consistency comparison rather than longitudinal analysis.*

Proof. (a) From **??**'s sample complexity: approximately n^* replication samples are needed to detect bias. Each T_{cycle} epoch accumulates roughly n^*/T_{cycle} effective samples (assuming uniform distribution). Total detection epochs $\approx n^* \cdot T_{\text{cycle}}/(\text{samples per epoch}) = O(T_{\text{cycle}} \cdot \log(2/\beta))$.

(b) From **??** and the weight update rule (**??**): malicious maintainer weights decay at an exponential rate; after C_{strike} consecutive flags, the weight drops to $O(e^{-\gamma C_{\text{strike}}})$.

(c) SCX Theorem 3's indistinguishability applies to longitudinal analysis of a single audit source. MetaAudit detects systematic bias by cross-sectional comparison of different audit sources, operating in an orthogonal information dimension (inter-auditor agreement), and thus does not contradict it. Formally: Theorem 3 states that $\forall$ algorithm $\mathcal{A}$ relying only on analytical properties of $\hat{S}_i(x)$, $\mathcal{A}$ cannot distinguish $H_0 : \epsilon = \epsilon_{\text{noise}}$ from $H_1 : \epsilon = \epsilon_{\text{difficulty}}$. But MetaAudit uses the statistic $D_n^{(i)}$, which depends on $\hat{S}_i(x) - \hat{S}_{\text{TPR}}(x)$—a **cross-source** difference—rather than single-source analysis; it operates outside Theorem 3's domain of definition. □

7 Bias Taxonomy and Extensions of Hoeffding Bound

7.1 Maintainer Bias Taxonomy

Definition 7.1 (Bias Types)**.** *Maintainer bias g_i can be classified as:*

B1 Constant Bias: *$g_i = c$ identical for all inputs x. Easiest to detect—$D_n^{(i)}$ converges to c on any input subset.*

B2 Directional Bias: *$g_i(x) = \mathbf{v}_i^\top \phi(x)$, where $\phi(x)$ are input features and $\mathbf{v}_i$ is the maintainer's preference vector. Detection requires sufficient coverage of the input space—if replication samples are orthogonal to the feature vector, the bias is hidden.*

B3 Conditional Bias: *$g_i(x) \neq 0$ only when x satisfies specific conditions (e.g., specific topics, specific sources). Detection requires stratified sampling covering all conditional regions.*

B4 Time-Varying Bias: *$g_i^{(t)}$ changes over time (e.g., slow poisoning $g_i^{(t)} = \delta \cdot t/T$). Requires cumulative detection over time windows or sliding-window tests.*

B5 Strategic Bias: *The maintainer is aware of meta-audit and behaves unbiasedly on replication samples ($g_i = 0$), while imposing bias on non-replication samples. Requires random audits so that the maintainer cannot distinguish replication samples from regular samples.*

7.2 Hoeffding Extension for Directional Bias

Theorem 7.2 (Directional Bias Detection)**.** *Let maintainer bias be of directional form $g_i(x) = \mathbf{v}_i^\top \phi(x)$, where $\phi : \mathcal{X} \to \mathbb{R}^d$ is a known feature map and $\mathbf{v}_i \in \mathbb{R}^d$ is an unknown bias vector. For n independent replication samples, let $\Phi_n = [\phi(x_1), \ldots, \phi(x_n)]^\top \in \mathbb{R}^{n\times d}$ be the design matrix. The OLS estimate of the bias vector $\mathbf{v}_i$ is:*

$$\hat{\mathbf{v}}_i = (\Phi_n^\top \Phi_n)^{-1} \Phi_n^\top \mathbf{D}_n^{(i)}, \tag{23}$$

where $\mathbf{D}_n^{(i)} = (D_1^{(i)}, \ldots, D_n^{(i)})^\top$ is the per-sample difference vector.

For any unit vector $\mathbf{u} \in \mathbb{R}^d$*, the linear combination* $\mathbf{u}^\top \hat{\mathbf{v}}_i$ *satisfies:*

$$\mathbb{P}\left(|\mathbf{u}^\top(\hat{\mathbf{v}}_i - \mathbf{v}_i)| \geq t\right) \leq 2\exp\left(-\frac{2nt^2}{R^2 \cdot \mathbf{u}^\top(\frac{1}{n}\Phi_n^\top \Phi_n)^{-1}\mathbf{u}}\right). \tag{24}$$

Proof. $\mathbf{u}^\top \hat{\mathbf{v}}_i = \mathbf{u}^\top(\Phi_n^\top \Phi_n)^{-1}\Phi_n^\top \mathbf{D}_n^{(i)} = \mathbf{a}^\top \mathbf{D}_n^{(i)}$, where $\mathbf{a} = \Phi_n(\Phi_n^\top \Phi_n)^{-1}\mathbf{u}$. This is a linear combination of n independent observations, each in $[-R, R]$. By the weighted Hoeffding inequality, the ℓ_2 norm squared of the coefficient vector $\mathbf{a}$ is $\|\mathbf{a}\|_2^2 = \mathbf{u}^\top(\Phi_n^\top \Phi_n)^{-1}\mathbf{u}$. The Hoeffding bound follows. □ □

7.3 Stratified Detection for Conditional Bias

Theorem 7.3 (Stratified Hoeffding Detection)**.** *Partition the input space* $\mathcal{X}$ *into* L *disjoint strata* $\mathcal{X}_1, \ldots, \mathcal{X}_L$*. For each stratum* ℓ*, independently draw* n_ℓ *replication samples. The per-stratum replication discrepancy statistic is* $D_{n_\ell}^{(i,\ell)}$*. Then the joint test:*

$$T_{strat} = \max_{\ell=1,\ldots,L} \frac{|D_{n_\ell}^{(i,\ell)}|}{\sqrt{R^2/(2n_\ell)}}. \tag{25}$$

Under the global null hypothesis $g_i(x) = 0, \forall x \in \mathcal{X}$*:*

$$\mathbb{P}\left(T_{strat} \geq t\right) \leq 2L \cdot \exp\left(-t^2\right). \tag{26}$$

Proof. Each stratum statistic $Z_\ell = D_{n_\ell}^{(i,\ell)}/\sqrt{R^2/(2n_\ell)}$ satisfies $\mathbb{P}(|Z_\ell| \geq t) \leq 2e^{-t^2}$ (by Hoeffding) under the null. The union bound (multiplying by L) yields the result. Bonferroni correction can obtain a tighter bound by controlling the family-wise error rate (FWER). □ □

7.4 Sliding Window for Time-Varying Bias

Theorem 7.4 (Sliding Window Hoeffding Detection)**.** *Let bias* $g_i^{(t)}$ *be approximately constant within a window of length* W*:* $|g_i^{(t)} - g_i^{(t')}| \leq \delta_W$ *for all* $|t - t'| \leq W$*. Accumulating* n_W *replication samples within the sliding window* $[t - W + 1, t]$*, define the window statistic:*

$$D_W^{(i,t)} = \frac{1}{n_W} \sum_{k \in window} \left(\hat{S}_i(x_k) - \hat{S}_{TPR}(x_k)\right). \tag{27}$$

Then:

$$\mathbb{P}\left(|D_W^{(i,t)} - \bar{g}_i^{(t)}| \geq t\right) \leq 2\exp\left(-\frac{2n_W t^2}{R^2}\right), \tag{28}$$

where $\bar{g}_i^{(t)} = \frac{1}{W}\sum_{\tau=t-W+1}^{t} g_i^{(\tau)}$ *is the average bias within the window.*

If $g_i^{(t)} = \delta \cdot t/T$ *(linear slow poisoning), then* $\bar{g}_i^{(t)} \approx \delta \cdot (t - W/2)/T$*. The window statistic can detect the average bias within the window; the detection condition is* $|\bar{g}_i^{(t)}| \geq \varepsilon$*, giving:*

$$t_{detect} \geq \frac{T \cdot \varepsilon}{\delta} + \frac{W}{2}. \tag{29}$$

Minimizing t_{detect} *involves a tradeoff between window length* W *(larger window* $\to$ *larger average bias, but also larger detection delay) and detection sensitivity. The optimal* W^* *satisfies* $W^* = \sqrt{2R^2\log(2/\alpha)/\delta^2}$*.*

Proof. The Hoeffding bound within the sliding window is direct—the n_W samples within the window are i.i.d. (assuming slowly varying bias), so **??** applies.

For linear poisoning $g_i^{(t)} = \delta \cdot t/T$, the window average is $\bar{g}_i^{(t)} = \frac{\delta}{T} \cdot \frac{1}{W} \sum_{\tau=t-W+1}^{t} \tau = \frac{\delta}{T} \cdot (t - \frac{W-1}{2})$. When $|\bar{g}_i^{(t)}| \geq \varepsilon$, it is statistically detectable, yielding $t \geq T\varepsilon/\delta + (W-1)/2 \approx T\varepsilon/\delta + W/2$.

Optimal window: detection delay $t_{\text{detect}} \approx T\varepsilon/\delta + W/2 + n_W$ (since n_W samples are needed). n_W is determined by $n_W \geq R^2 \log(2/\alpha)/(2\varepsilon^2)$. Total delay $\approx T\varepsilon/\delta + W/2 + R^2 \log(2/\alpha)/(2\varepsilon^2)$. Differentiating w.r.t. W (noting n_W is typically proportional to W, assuming constant sampling rate) yields the optimal W. □ □

8 Strategic Bias and Random Audits

8.1 Audit Game Model

Definition 8.1 (Audit Game)**.** *The game between maintainer M_i and the Meta-Auditor (MA):*

- ***Maintainer action:*** *For each input x, choose whether to impose bias $g_i(x) \in \{0, b\}$ ($b > 0$ is the bias magnitude). Imposing bias yields payoff $\rho \cdot b$ (e.g., advancing a specific agenda), but carries detection risk.*
- ***Meta-Auditor action:*** *Choose replication probability $p \in (0, 1]$, i.e., with probability p, independently replicate any audit output (without informing the maintainer which outputs are selected).*
- ***Detection:*** *If a biased input is replicated, it is detected with probability $q(n, \varepsilon, \alpha)$, where q is given by* **??***.*
- ***Punishment:*** *If bias is detected, the maintainer is removed, incurring loss L (discounted present value of all future payoffs).*

Theorem 8.2 (Deterrence Effect of Random Audits)**.** *In the audit game, if the replication probability p satisfies:*

$$p \cdot q \cdot L > \rho \cdot b, \tag{30}$$

then the maintainer's dominant strategy is to impose no bias ($g_i = 0$). The minimum deterrence value of p is:

$$p_{\min} = \frac{\rho \cdot b}{q \cdot L}. \tag{31}$$

Proof. The expected payoff of a single bias imposition by the maintainer is $\rho b - p \cdot q \cdot L$ (gain minus expected punishment). If this is negative, i.e., $\rho b < p \cdot q \cdot L$, the expected net payoff of imposing bias is negative, and a rational maintainer does not impose bias. Note: this is a single-shot analysis. In the repeated game, the cost of being detected also includes the loss of all future-period payoffs, so L is larger in the repeated setting, making the deterrence condition easier to satisfy. □ □

Corollary 8.3 (Infinite-Horizon Deterrence)**.** *In an infinite-horizon repeated game with discount factor $\beta \in (0, 1)$, the maintainer is permanently removed upon detection. The expected long-term payoff of imposing bias is:*

$$\rho b - p \cdot q \cdot \frac{\rho \cdot \bar{S}}{1 - \beta},$$

where $\bar{S}$ is the maintainer's average legitimate payoff per period. The deterrence condition is:

$$p \cdot q > \frac{\rho b(1 - \beta)}{\rho \cdot \bar{S}}. \tag{32}$$

*Since $\bar{S} \gg b$ (legitimate audit payoff is much larger than bias payoff), the right-hand side is extremely small, so p can be extremely small—**long-tenure maintainers are easier to deter than short-tenure maintainers**.*

8.2 Indistinguishable Replication

Definition 8.4 (Indistinguishable Replication)**.** *The probability distribution of replication samples is identical to that of regular audit samples: if a maintainer observes input x, they cannot determine, with probability better than random guessing, whether x is a replication sample. Formally, the replication flag $R_x \in \{0, 1\}$ is independent of any observable feature of x:*

$$\mathbb{P}(R_x = 1 \mid x) = \mathbb{P}(R_x = 1) = p, \quad R_x \perp\!\!\!\perp \phi(x), \tag{33}$$

where $\phi(x)$ is any input feature observable by the maintainer.

Remark 8.5. *Indistinguishability is the* ***core security property*** *of random audits. Without it, a maintainer could strategically impose bias only on non-replication samples—this is equivalent to a special case of directional bias: $g_i(x) = b \cdot \mathbf{1}_{R_x=0}$. In practice, indistinguishability is achieved by: (a) replicators drawing samples directly from public audit logs, rather than using an independent sampling mechanism; (b) sufficiently large replication delay so that maintainers cannot correlate replication requests with specific trigger events; (c) using cryptographic commitments to ensure audit samples were selected but hidden from the maintainer before they produce their output.*

9 Numerical Analysis and Simulation

9.1 Detection Sample Complexity Table

Table 1: Required Replication Sample Size n under Hoeffding Detection

α	$1-\beta$	ε	R	$n_{\text{Hoeffding}}$	$n_{\text{Bernstein}}$ (if $\sigma = 0.1R$)
0.05	0.80	0.10	1.0	614	152
0.05	0.95	0.10	1.0	1,082	268
0.01	0.95	0.10	1.0	1,498	371
0.05	0.95	0.05	1.0	4,328	1,072
0.05	0.95	0.01	1.0	108,200	26,800
0.01	0.99	0.10	1.0	2,126	527
0.01	0.99	0.05	1.0	8,504	2,108

9.2 Rotation Cumulative Bias Simulation

Table 2: Cumulative Bias Comparison under Rotation ($G_{\max} = 0.2$, $T = 100$)

N	$L_{\max}$	$C_{\min}$	Max Cumulative Bias	Reduction vs. No Rotation
No rotation	—	—	20.0	—
3	1	2	6.67	66.7%
5	1	4	4.00	80.0%
5	2	3	8.00	60.0%
7	1	6	2.86	85.7%
10	2	8	4.00	80.0%

10 Integration with the SCX Framework

10.1 Reflexivity as the Ninth SCX Principle

The SCX Equality Framework's core axioms (Theorems 1–8) construct a complete theoretical edifice from expert consensus detection to human-AI collaborative auditing. However, these theorems all assume auditor reliability. The meta-audit protocol fills this reflexivity gap and can be regarded as SCX's **Ninth Principle: the Principle of Reflexivity**:

> **Principle 9 (Reflexivity Verification).** In any system claiming to implement SCX auditing, the auditors themselves must be subject to audit at the same or higher level. This meta-audit must: (a) make all audit logs public; (b) permit and encourage independent third-party random replication; (c) implement maintainer rotation and multi-maintainer consensus; (d) use statistical tests (such as the Hoeffding bound) to quantitatively detect auditor bias, with test power and sample complexity declared in advance and publicly verified.

10.2 Detailed Consistency Analysis with Theorem 3

Theorem 10.1 (Consistency of Meta-Audit with Theorem 3)**.** *The Meta-Audit protocol* MetaAudit *does not violate SCX Theorem 3 (noise-difficulty indistinguishability). Specifically:*

(a) MetaAudit*'s detection statistic $D_n^{(i)} = \frac{1}{n}\sum_k(\hat{S}_i(x_k) - \hat{S}_{TPR}(x_k))$ involves the output difference of* ***two independent audit sources****, not the internal analysis of a single source;*

(b) Theorem 3's indistinguishability statement applies to observables of a single audit source $\{\hat{S}(x), \nabla\hat{S}(x), H_{\hat{S}}(x)\}$;

(c) $D_n^{(i)}$ derives its information from inter-auditor ***disagreement****, an information dimension outside Theorem 3's domain;*

(d) Formally: if Theorem 3 states that $\forall\mathcal{A}_{internal}$ relying only on the internal structure of $\hat{S}_i(x)$ (gradient, Hessian, etc.), $\mathcal{A}_{internal}$ cannot distinguish H_0 from H_1, then MetaAudit *falls outside this statement's scope, because* MetaAudit*'s core input is $\hat{S}_i(x) - \hat{S}_j(x)$ (cross-source difference), which cannot be represented in Theorem 3's input space.*

11 Implementation Considerations

11.1 Cryptographic Guarantees for Audit Logs

CL1 Content-Addressable Storage. Each audit entry $(x, \hat{S}_i(x), t)$ is uniquely identified by its hash $\mathsf{hash} = H(x\|\hat{S}_i(x)\|t)$. Audit logs are stored in a content-addressable system (e.g., IPFS), ensuring entries cannot be tampered with.

CL2 Merkle Chain. Log entries form a hash chain: $\mathsf{hash}_j = H(x_j\|\hat{S}_i(x_j)\|t_j\|\mathsf{hash}_{j-1})$. Any modification to an entry changes all subsequent hashes, making tampering publicly detectable.

CL3 Timestamping. Periodically anchor the latest hash to a public blockchain (e.g., via Opentimestamps or similar mechanisms), providing non-repudiable temporal proof.

CL4 ZK Replication Proofs. A TPR can generate a zero-knowledge proof π proving that $\exists$ replication computation $\hat{S}_{\text{TPR}}(x) = y$ and the maintainer's logged entry $\hat{S}_i(x) = y'$ satisfy $|y - y'| \geq \tau$ (significant discrepancy), without revealing the TPR's computation method or intermediate results—protecting the TPR's audit methodology intellectual property.

11.2 Maintainer Pool Governance

GOV1 Admission. New maintainers must pass: a mathematical competency test (ensuring correct computation of Cercis Score and gradient-field diagnostics), a bias declaration (disclosing any conflicts of interest that could affect auditing), and a stake deposit (economic collateral forfeited if bias is detected).

GOV2 Exit. Maintainers may voluntarily exit (with advance notice and completion of the current audit cycle). Forced exit triggers automatically after C_{strike} consecutive bias flags.

GOV3 Reputation System. Each maintainer's historical performance is publicly recorded: bias detection count, replication consistency score, consensus contribution rate. Reputation affects weight w_i and maintainer scheduling priority.

GOV4 Decentralized Coordination. Rotation scheduling and replicator assignment are determined by an on-chain random beacon (e.g., RANDAO) or verifiable random function (VRF), preventing the schedule itself from being manipulated.

12 Related Work

The concept of meta-audit intersects with the following areas:

1. **Statistical Audit Sampling.** Attribute and variable sampling methods in financial auditing share similarities with our replication statistics, but traditional audit sample-size calculations rely on normal approximations rather than Hoeffding bounds and do not involve inter-maintainer game theory.

2. **Distributed Systems Byzantine Fault Tolerance (BFT).** PBFT (Castro & Liskov, 1999), Tendermint (Buchman, 2016), HotStuff (Yin et al., 2019), and other consensus protocols handle Byzantine fault tolerance for general state-machine replication, whereas this paper focuses on numerical consensus of audit scores, enabling more efficient statistical aggregation (e.g., weighted median).

3. **Debiasing Peer Review.** NeurIPS experiments (2014, 2022), ICML's randomized reviewer assignments, and bid calibration methods attempt to reduce reviewer bias, but all rely on institutional design rather than statistical detection guarantees—the meta-audit protocol provides a mathematically more rigorous approach.

4. **Prediction Markets & Truth Discovery.** Bayesian Truth Serum (Prelec, 2004) and weighted majority algorithms share mathematical parallels with our weighted consensus mechanism, but our weight updates are based on statistical hypothesis testing rather than subjective scoring.

5. **Algorithmic Auditing.** Fairness audits and model cards focus on algorithmic output bias, whereas this paper focuses on the bias of the auditor itself—auditing at the meta-level.

13 Open Problems and Future Work

OP1 Minimum Variance Unbiased Estimation (MVUE). The current Hoeffding bound relies on the boundedness assumption. Do tighter distribution-free bounds exist (e.g., using the empirical Bernstein inequality) that exploit sample variance information while remaining distribution-free? When $\sigma_i^2 \ll R^2$, the efficiency gain is substantial.

OP2 Adaptive Detection Thresholds. $\tau = \varepsilon/2$ is optimal for symmetric allocation. Do data-driven adaptive thresholds exist (e.g., Lawless & Singhal's hybrid stopping rules) that reduce detection delay without increasing false positives?

OP3 MPC Meta-Audit. Can the MetaAudit protocol be implemented as secure multi-party computation (MPC), enabling bias detection without revealing any individual auditor's raw audit outputs? This is compatible with the federated audit framework of the FA-Theorem.

OP4 Fully Autonomous Meta-Audit. When both auditors and meta-auditors are AI systems, how is the recursion depth of meta-audit chosen? Is there a natural stopping point (e.g., reaching the limit of available computational resources, or recursive benefit diminishing below a threshold)?

OP5 Meta-Meta-Audit. Who audits the meta-auditor? This paper does not fully resolve this question—we assume TPR honesty or majority consensus reliability. A fully recursive solution requires infinitely many levels of audit, but in practice 2–3 levels provide effective deterrence, because each additional level exponentially increases the operational complexity for the attacker.

OP6 Game-Theoretic Completeness. The audit game **??** only analyzes the extreme cases of single-shot and infinite-horizon repeated play. Finite-horizon, incomplete-information, and multi-maintainer collusion scenarios require more refined analysis. In particular: when $b \geq 2$ maintainers simultaneously impose bias, what is the reliability condition for k-of-n consensus?

14 Conclusion

This paper rigorously formalizes the meta-audit protocol within the SCX Equality Framework, answering the reflexive question "who audits the auditor." Core results include:

- Formalization of the maintainer bias model $\hat{S}_i(x) = S(x) + g_i + \epsilon_i(x)$, distinguishing systematic bias g_i from random error $\epsilon_i(x)$.
- Proof, via Hoeffding's inequality, that detecting $|g_i| \geq \varepsilon$ at significance level α and power $1-\beta$ requires a lower bound of $n \geq \frac{R^2}{2\varepsilon^2}(\sqrt{\log(2/\alpha)} + \sqrt{\log(2/\beta)})^2$ independent replication samples.
- Hoeffding extension theorems for multi-TPR joint detection, directional bias, conditional bias, and time-varying bias (slow poisoning), covering all major bias types.
- Cumulative bias upper bound $O(G_{\max} \cdot T \cdot L_{\max}/(L_{\max} + C_{\min}))$ for maintainer rotation scheduling, achieving 66.7%–85.7% cumulative bias reduction under optimal parameters ($L_{\max} = 1, C_{\min} = N - 1$).
- Reliability condition $b < \min(k, n-k+1)$ for k-of-n Byzantine consensus, and Hoeffding error bound for weighted consensus.
- Deterrence condition $p \cdot q \cdot L > \rho \cdot b$ for random audit games, and the security property of indistinguishable replication.
- Consistency proof with SCX Theorem 3—meta-audit operates on cross-source disagreement information, not violating noise/difficulty indistinguishability.

The meta-audit protocol elevates the SCX Framework from "trust the auditor" to "verify the auditor," achieving reflexive completeness of the audit regime. It is both a mathematical theorem and an institutional design—combining statistical rigor with game-theoretic incentive compatibility.

Appendix A: Key Notation

Table 3: Key Notation

Symbol	**Meaning**
$S(x)$	True Cercis Score
$\hat{S}_i(x)$	Auditor i's audit output
g_i	Systematic bias of maintainer i
$\epsilon_i(x)$	Random error of maintainer i
σ_i^2	Random error variance
R	Audit range width $S_{\max} - S_{\min}$
$D_n^{(i)}$	Replication discrepancy statistic
n	Replication sample size
α	Significance level
$1 - \beta$	Test power
ε	Minimum detectable bias
τ	Detection threshold
K	Number of TPR replicators
N	Maintainer pool size
$L_{\max}$	Maximum consecutive tenure
$C_{\min}$	Minimum cool-down period
b	Number of Byzantine maintainers
p	Random replication probability
w_i	Maintainer trust weight
$W_{\mathcal{H}}$	Sum of honest maintainer weights

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